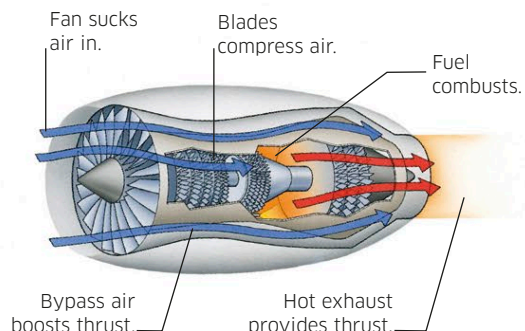
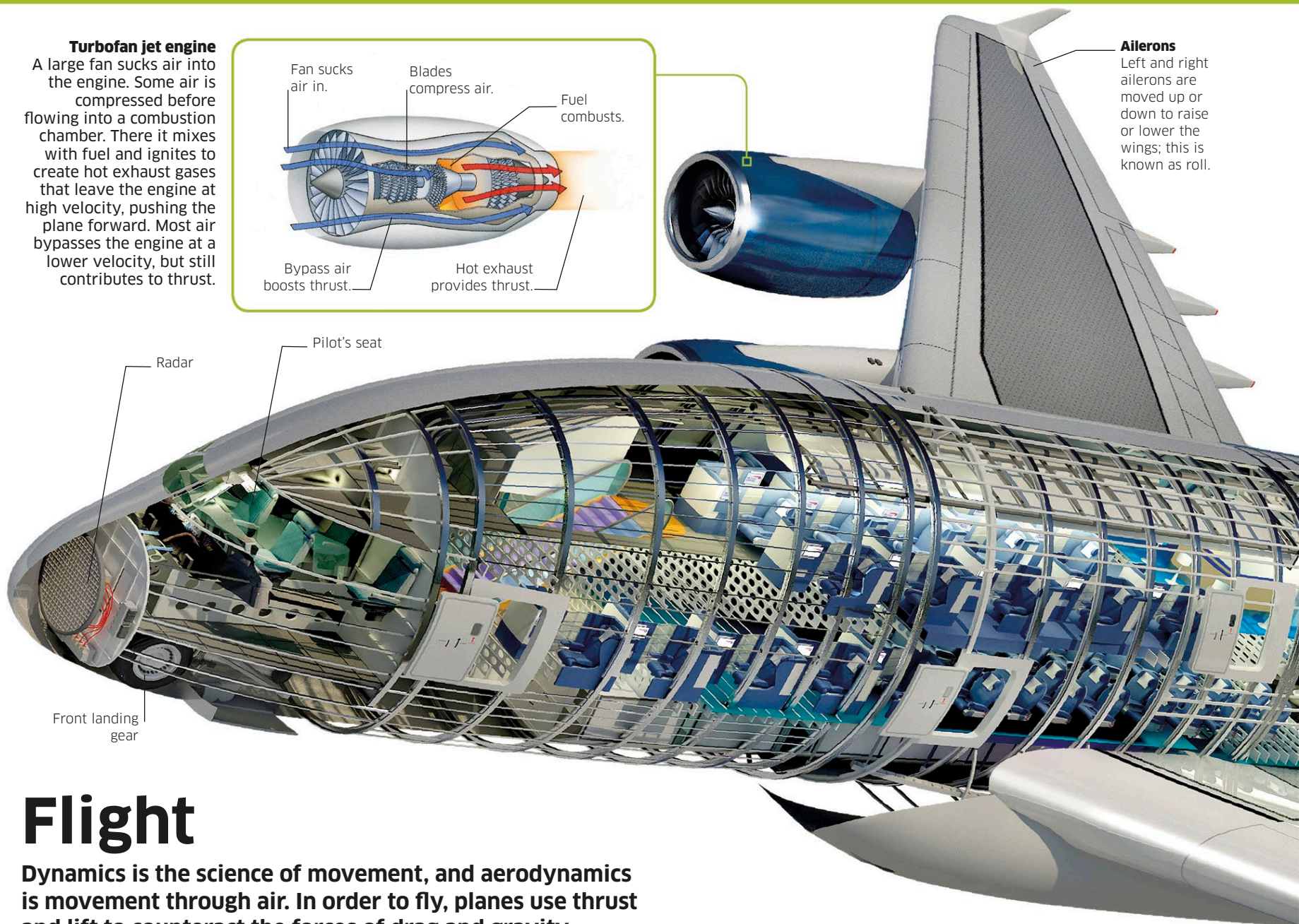


**Turbofan jet engine**

A large fan sucks air into the engine. Some air is compressed before flowing into a combustion chamber. There it mixes with fuel and ignites to create hot exhaust gases that leave the engine at high velocity, pushing the plane forward. Most air bypasses the engine at a lower velocity, but still contributes to thrust.

**Ailerons**

Left and right ailerons are moved up or down to raise or lower the wings; this is known as roll.



# Flight

**Dynamics is the science of movement, and aerodynamics is movement through air. In order to fly, planes use thrust and lift to counteract the forces of drag and gravity.**

Just over a hundred years since the first powered flight, today more than 100,000 planes fly every day and it seems normal to us that an airliner weighing as much as 619 tons when laden can take to the skies. To take off, a plane must generate enough lift to overcome gravity, using the power of its engines to create drag-defying thrust.

**Airbus A380**

The Airbus 380 is the world's biggest passenger aircraft: 234 ft (73 m) long with a wing span of 262 ft (79.8 m), it can seat 555 people on two decks and carry 165 tons of cargo.

**The forces of flight**

Four forces act upon an airplane traveling through the air: thrust, lift, gravity, and drag. Thrust from the engines pushes the plane forward, forcing air over the wings, which creates lift to get it off the ground, while gravity pulls the plane downward, and drag – or air resistance – pulls it backward. In level flight at a constant speed, all four of these forces are perfectly balanced.

